

Adding and Subtracting Fractions

Answer Key

Grade 4–5

Quick Answers

- | | |
|---|--|
| 1. $\frac{5}{6}$ | 7. 2 |
| 2. second addend = $1 \cdot \frac{1}{4}$, third addend = $1 \cdot \frac{1}{4}$ | 8. $\frac{7}{8}$ L |
| 3. $\frac{3}{8}$ | 9. fraction planted with something other than beans = $1 \cdot \frac{1}{3}$, — |
| 4. $\frac{7}{10}$ mi | 10. (a) fraction of the bottle Maya drank = $\frac{3}{4}$,
(b) fraction of the bottle left = $1 \cdot \frac{1}{4}$ |
| 5. $1 \cdot \frac{1}{2}$ | |
| 6. $\frac{3}{10}$ | |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 4–5

4.NF.B.3 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 13 tagged checks.

- 1.** The bar is cut into sixths; the dark part shows $\frac{2}{6}$ and the light part shows $\frac{3}{6}$. Add: $\frac{2}{6} + \frac{3}{6}$

Solution: The pieces are all sixths — the same size — so we can just count them: 2 dark pieces plus 3 light pieces makes 5 shaded pieces out of 6. The denominator stays 6.

$$\frac{2}{6} + \frac{3}{6} = \boxed{\frac{5}{6}}$$

- 2.** Fill in the two missing unit fractions: $\frac{3}{4} = \frac{1}{4} + \square + \square$

Solution: Three fourths is three copies of one fourth: $\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$. Each missing addend is $\boxed{\frac{1}{4}}$ and again $\boxed{\frac{1}{4}}$

- 3.** Subtract: $\frac{5}{8} - \frac{2}{8}$

Solution: Same-size pieces (eighths), so subtract the counts: $5 - 2 = 3$ pieces, each an eighth. $\frac{5}{8} - \frac{2}{8} = \boxed{\frac{3}{8}}$

- 4.** Ella walks $\frac{3}{10}$ mi to the park, then $\frac{4}{10}$ mi more to school. How far in all?

Solution: Both distances are in tenths of a mile, so add the counts of tenths: $3 + 4 = 7$ tenths. $\frac{3}{10} + \frac{4}{10} = \frac{7}{10}$. Total distance: $\boxed{\frac{7}{10} \text{ mi}}$

5. The bar shows $\frac{7}{8}$ shaded. Cross out $\frac{3}{8}$, then subtract: $\frac{7}{8} - \frac{3}{8}$

Solution: Crossing out 3 of the 7 shaded eighths leaves $7 - 3 = 4$ shaded eighths: $\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$. Four eighths is half the bar, so simplify by dividing top and bottom by 4: $\frac{4}{8} = \boxed{\frac{1}{2}}$

6. Find the missing fraction: $\square + \frac{4}{10} = \frac{7}{10}$

Solution: Think of it as a missing-addend problem: how many tenths must be added to 4 tenths to reach 7 tenths? Since $7 - 4 = 3$, the missing fraction is $\boxed{\frac{3}{10}}$

7. Fill in the missing numerator: $\frac{5}{6} = \frac{3}{6} + \frac{\square}{6}$

Solution: Decompose 5 sixths as 3 sixths plus the rest: $5 - 3 = 2$ sixths are missing. The missing numerator is $\boxed{2}$

8. Ravi mixes $\frac{3}{8}$ L of red, $\frac{3}{8}$ L of blue, and $\frac{1}{8}$ L of white paint. How many liters in all?

Solution: All three amounts are eighths of a liter, so add the numerators: $3 + 3 + 1 = 7$ eighths. $\frac{3}{8} + \frac{3}{8} + \frac{1}{8} = \frac{7}{8}$. Total paint: $\boxed{\frac{7}{8} \text{ L}}$

9. A garden has 12 equal sections; $\frac{9}{12}$ is planted and beans fill $\frac{5}{12}$. What fraction is planted with something other than beans? Explain why the denominator stays 12.

Solution: Planted minus beans: $\frac{9}{12} - \frac{5}{12} = \frac{4}{12}$. Simplify by dividing top and bottom by 4: $\frac{4}{12} = \frac{1}{3}$. Fraction planted with something other than beans: $\boxed{\frac{1}{3}}$

Explanation (instructor judges): a correct explanation says the garden is still cut into twelfths — subtracting removes pieces but does not change the size of each piece, so only the numerator (the count of pieces) changes.

10. Maya drinks $\frac{1}{8}$ of her bottle in the morning and $\frac{5}{8}$ at lunch. (a) What fraction has she drunk? (b) What fraction is left?

Solution: (a) Add the two drinks: $\frac{1}{8} + \frac{5}{8} = \frac{6}{8}$, and $\frac{6}{8}$ simplifies (divide top and bottom by 2) to $\boxed{\frac{3}{4}}$

(b) The whole bottle is $\frac{8}{8}$. Left over: $\frac{8}{8} - \frac{6}{8} = \frac{2}{8}$, which simplifies to $\boxed{\frac{1}{4}}$